4.0 Chargement uniaxial

GMC-2001
Résistance des matériaux

Plan

1. Introduction
2. Barre avec déformation axiale uniforme
3. Barre avec déformation axiale non uniforme
4. Système hyperstatique
5. Contraintes thermiques
6. Conclusions
7. Problèmes

Hibbeler: Sections 4.1, 4.2, 4.4, 4.6.

1. Introduction

Le chargement uniaxial est un chargement simple qui se retrouve dans beaucoup d'applications.



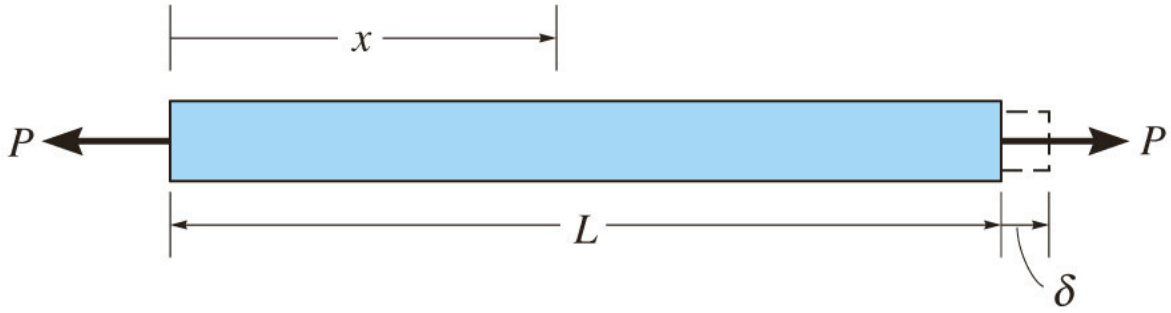


2. Barre avec déformation axiale uniforme

Une barre qui a:

1) Une section *A* constante
2) Un module élastique *E* constant
3) Des forces axiales *P* appliquées à ses extrémités.

subit une ***déformation axiale uniforme***.



$$\sigma_x = \frac{P}{A}$$

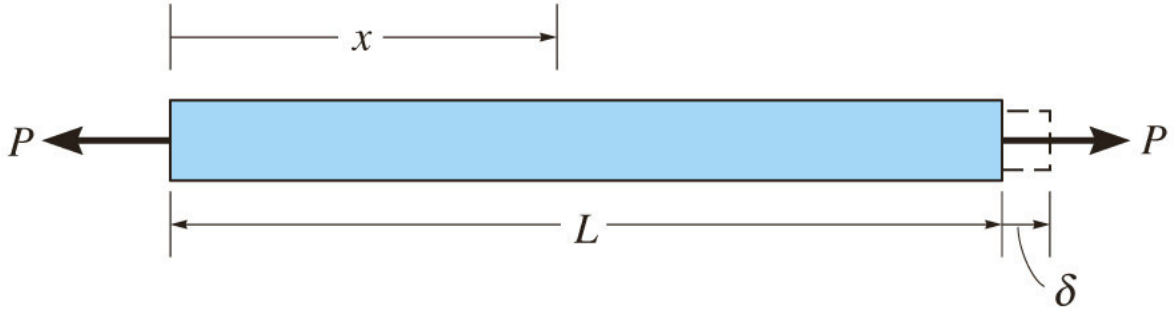
σ_x est uniforme le long de la barre.

$$\varepsilon_x = \frac{\sigma_x}{E} = \frac{P}{AE}$$

ε_x est uniforme le long de la barre.

2. Barre avec déformation axiale uniforme

Quelle est l'élongation δ de la barre?



$$\varepsilon_x = \frac{P}{AE} \quad \text{ou} \quad \varepsilon_x = \frac{\delta}{L}$$

Comme la déformation ε_x est constante le long de la barre, l'élongation de la barre est égale à:

$$\delta = \varepsilon_x L = \frac{P}{AE} L$$

$$\boxed{\delta = \frac{PL}{AE}}$$

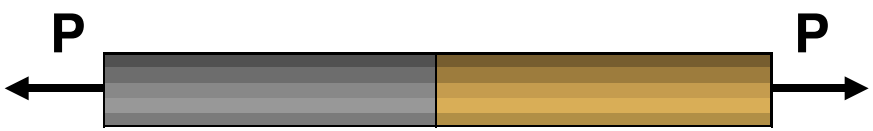
Relation force-déplacement

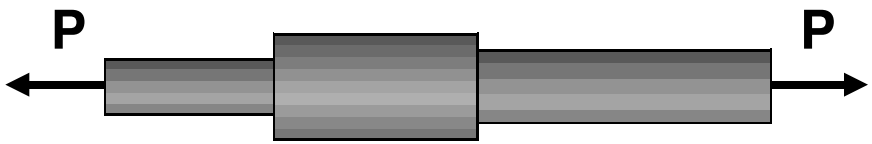
Il est **important** de vous rappeler les hypothèses qui ont été posées quand vous utilisez $\delta = PL/AE$.

2. Barre avec déformation axiale uniforme

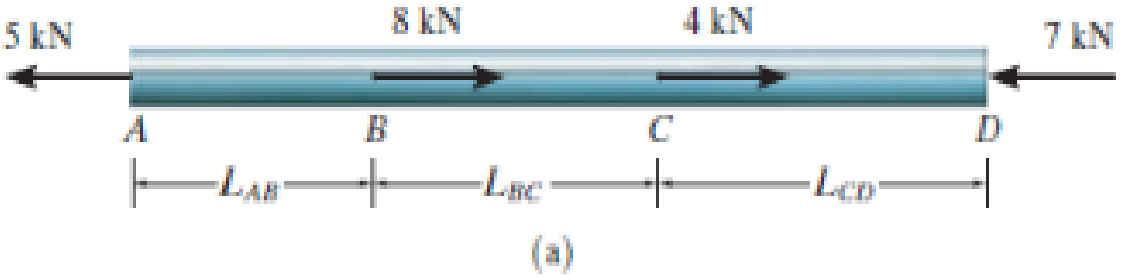
Note:

Pour les cas où la barre est composée de segments faits de matériaux ou de sections différentes ou qu'elle est soumise à plusieurs forces axiales, les déformations seront uniformes dans chaque segment ou *P*, *A* et *E* sont constants.





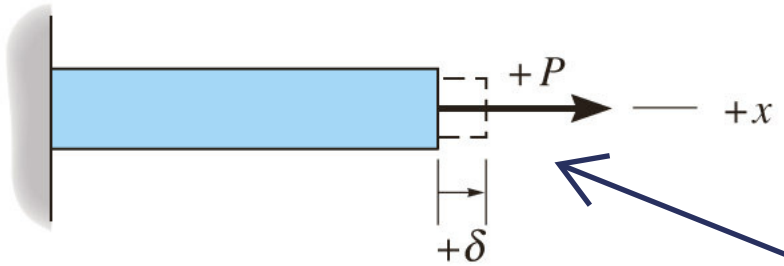
$$\delta = \Sigma \frac{PL}{AE}$$

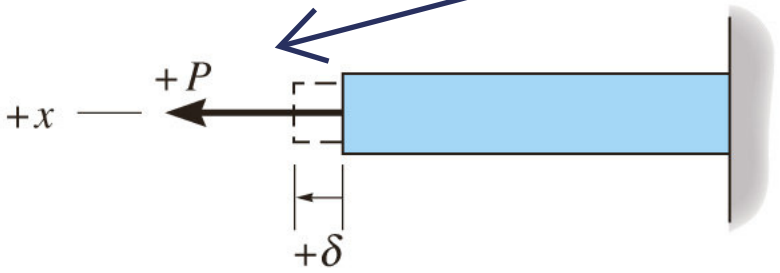


2. Barre avec déformation axiale uniforme

Convention de signe:

$$\delta = \frac{PL}{AE} \qquad \delta = \Sigma \frac{PL}{AE}$$





Barre en traction:

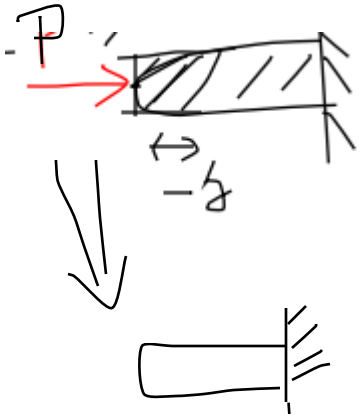
P > 0

δ > 0

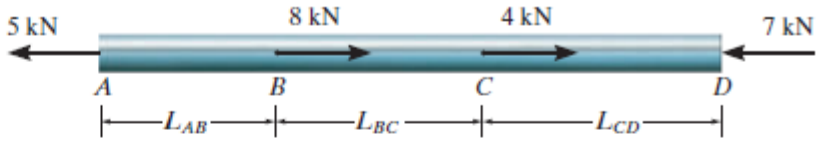
Barre en compression:

P < 0

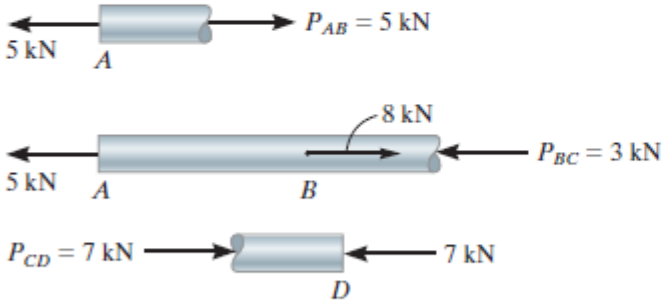
δ < 0



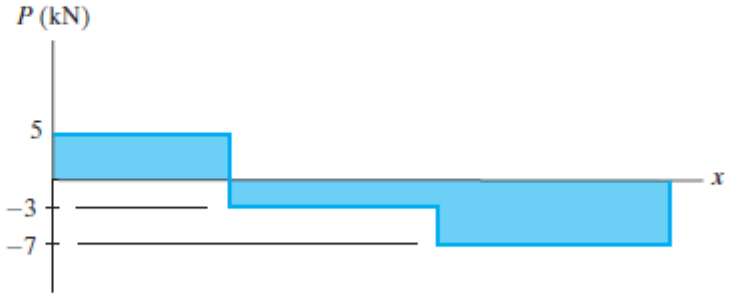
2. Barre avec déformation axiale uniforme



(a)



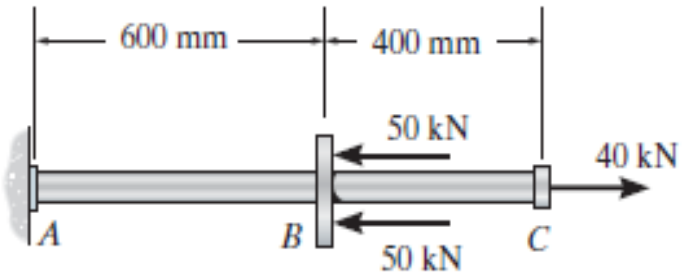
(b)



(c)

- Exemple :

F4–1. The 20-mm-diameter A-36 steel rod is subjected to the axial forces shown. Determine the displacement of end C with respect to the fixed support at A.



F4–1

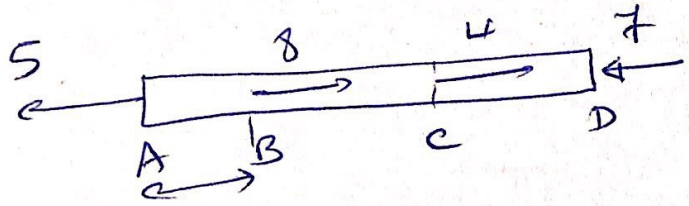
$$A = \frac{\pi}{4}(0.02^2) = 0.1(10^{-3})\pi \text{ m}^2$$

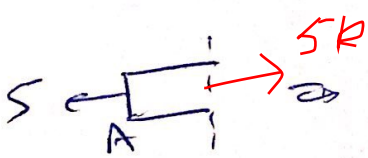
$$\delta_C = \frac{1}{AE}\{40(10^3)(400) + [-60(10^3)(600)]\}$$

$$= \frac{-20(10^6) \text{ N} \cdot \text{mm}}{AE}$$

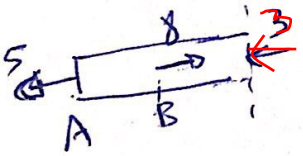
$$= -0.318 \text{ mm}$$

Exemple: $A = cte$, $E = cte$, $P \neq cte$, $L = \sum L_i$ → P_i

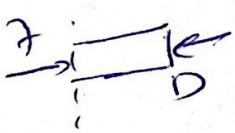




AB: $P_{AB} = 5\,kN \Rightarrow \delta_{AB} = \dfrac{+5.10^3 . L_{AB}}{AE}$



BC: $P_{BC} = -3\,kN \Rightarrow \delta_{BC} = \dfrac{-3.10^3 L_{BC}}{AE}$

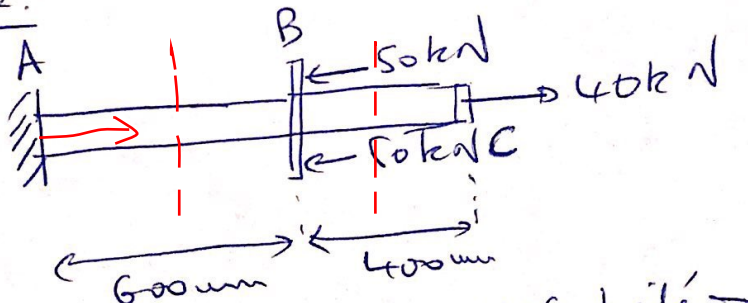


CD: $P_{CD} = -7\,kN \Rightarrow \delta_{CD} = \dfrac{-7.10^3 L_{CD}}{AE}$

$$\delta = \delta_{AB} + \delta_{BC} + \delta_{CD} = \frac{(5L_{AB} - 3L_{BC} - 7L_{CD})10^3}{AE}$$

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Exemple :

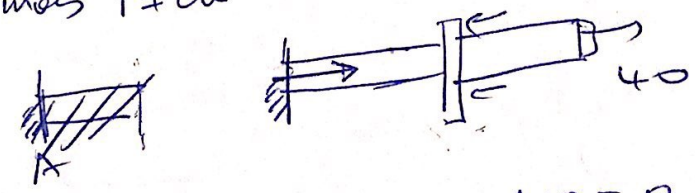




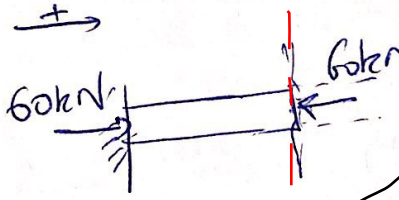
Même section, même module d'Elasticité $\Rightarrow$ $A = cte$, $E = cte$

mais $P \neq cte$

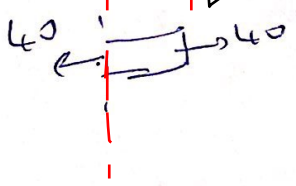
$$\delta = \frac{PL}{AE}$$



$\Sigma F_x \rightarrow +$: $F_A + 40 - 100 = 0 \Rightarrow F_A = 100 - 40 = 60\,kN$



$$\delta_{AB} = \frac{-60.10^3 \; 600.10^{-3}}{\left(\frac{20.10^{-3}}{2}\right)^2 \pi \; 200.10^9} = -0,57\,mm$$



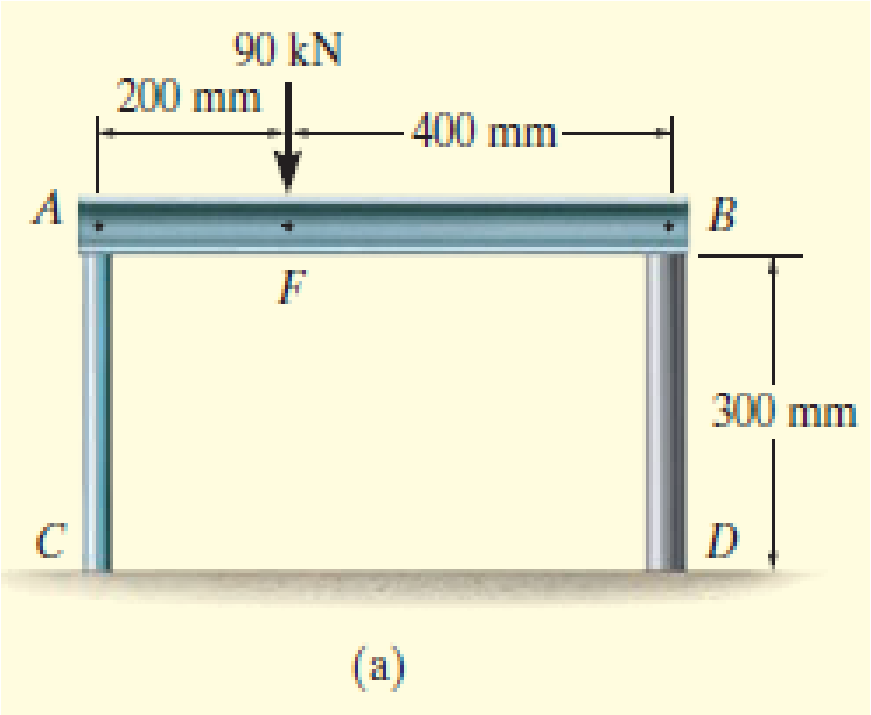
$$\delta_{BC} = \frac{40.10^3 \; 400.10^{-3}}{\left(\frac{20.10^{-3}}{2}\right)^2 \pi \; 200.10^9} = 0,25\,mm$$

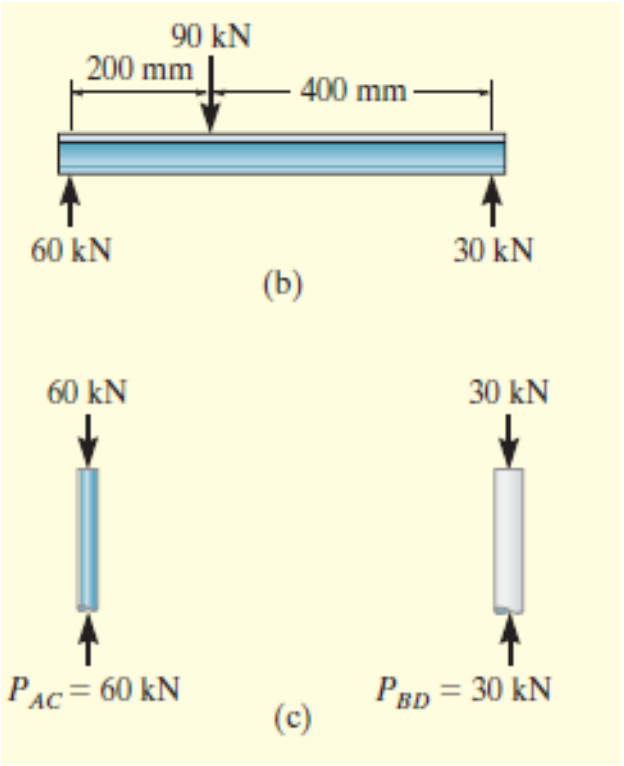
$$\delta = \delta_{AB} + \delta_{BC} = -0,57 + 0,25 = -0,32\,mm$$

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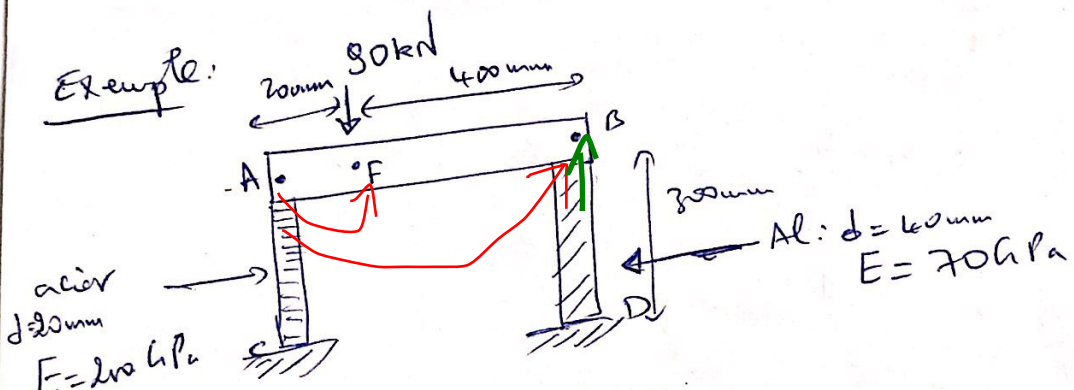
2. Barre avec déformation axiale uniforme

Rigid beam AB rests on the two short posts shown in Fig. 4–8*a*. AC is made of steel and has a diameter of 20 mm, and BD is made of aluminum and has a diameter of 40 mm. Determine the displacement of point F on AB if a vertical load of 90 kN is applied over this point. Take $E_{st} = 200$ GPa, $E_{al} = 70$ GPa.





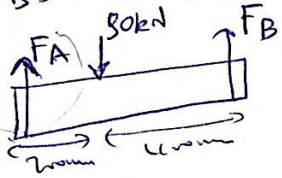
Exemple :





$\delta_{AC} = ?$ $\delta_{BD} = ? \Rightarrow \delta_F = ?$

D.C.L (AB) ?



$\vec{\mathcal{M}}_A = \vec{0} \Rightarrow -200.10^{-3} \times 90.10^3 + 600\times10^{-3} \times F_B = 0$

$\Rightarrow \frac{+200\times90}{600.10^{-3}} = F_B \Rightarrow \boxed{F_B = \frac{18000}{600.10^{-3}} = 30kN}$

$F_A + F_B - 90kN = 0$

$\Rightarrow \boxed{F_A = 60kN}$

D.C.L (AC)



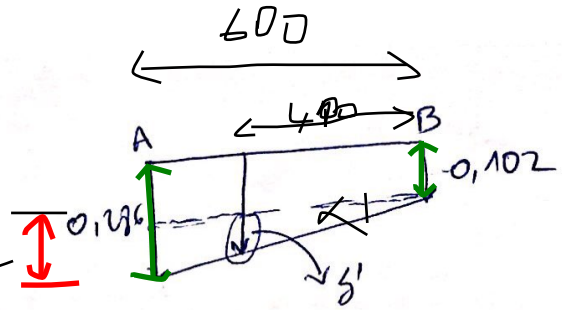
$\delta_{AC} = \frac{-60.10^3 . 300.10^{-3}}{200.10^9 \times \left(\frac{20.10^{-3}}{2}\right)^2 \pi} = -0,286\ mm$



D.C.L (BD)

$\delta_{BD} = \frac{-30.10^3 . 300.10^{-3}}{\left(\frac{40.10^{-3}}{2}\right)^2 \pi\ 70.10^9} = -0,102\ mm$

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(0,286 − 0,172)

(3)

$\delta_F = 0,102 + \delta'$

$tg\,2 = \frac{0,286 - 0,102}{600} = \frac{\delta'}{400}$

$\Rightarrow \delta' = \frac{0,286 - 0,102}{600} \times 400 = 0,122$

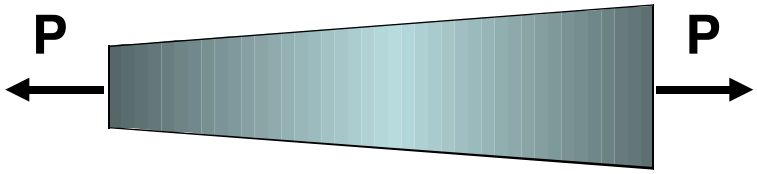
$\delta_F = 0,224$ mm.

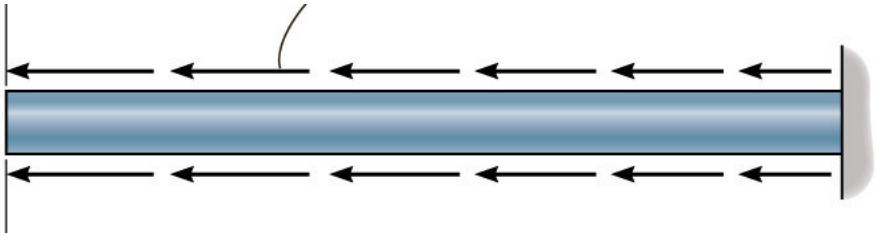
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3. Barre avec déformation axiale non uniforme

- Lorsque:
  - La barre a une section variable ou
  - Le chargement varie le long de la barre,

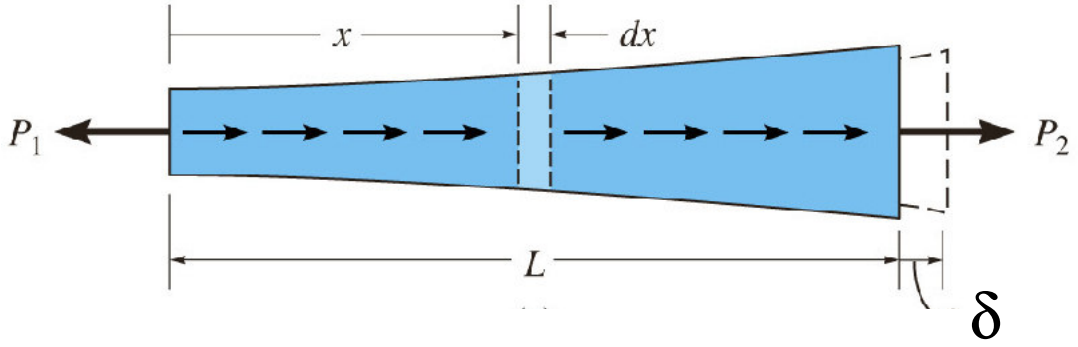
la contrainte et la déformation varient de façon continue le long de la barre.

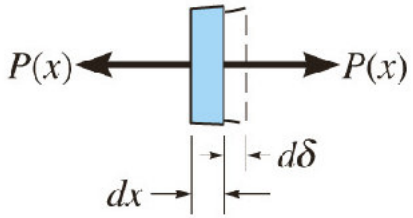




3. Barre avec déformation axiale non uniforme

- **Cas général**:





$$\sigma_x = \frac{P(x)}{A(x)}$$

La contrainte σ_x varie en fonction de x.

$$\varepsilon_x = \frac{\sigma_x}{E} = \frac{P(x)}{EA(x)} = \frac{d\delta}{dx} \Rightarrow d\delta = \frac{P(x)dx}{A(x)E}$$

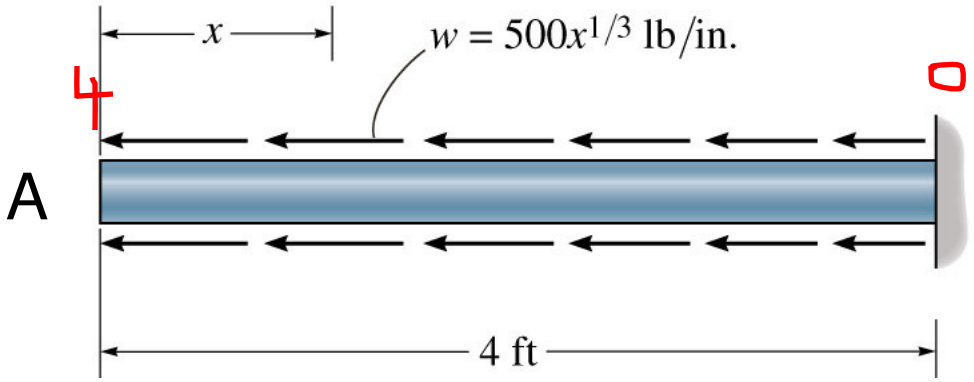
La déformation ε_x varie en fonction de x. ⇒

$$\delta = \int_0^L \frac{P(x)dx}{A(x)E}$$

3. Barre avec déformation axiale non uniforme

- **Problème 4-6:**

Soit une barre de section A=3 in² et de module E=35 Msi.
Déterminer le déplacement de l'extrémité A lorsque la barre est soumise au chargement réparti w(x).

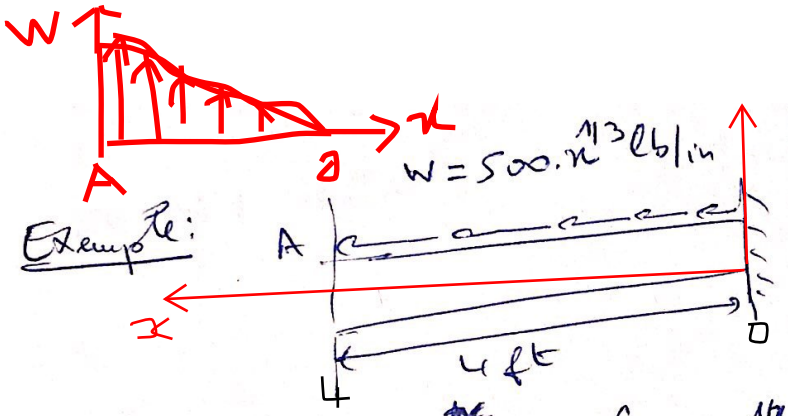


$$P(x) = \int_0^x w\,dx = 500\int_0^x x^{\frac{1}{3}}\,dx = \frac{1500}{4}x^{\frac{4}{3}}$$

$$\delta_A = \int_0^L \frac{P(x)\,dx}{AE} = \frac{1}{(3)(35)(10^6)}\int_0^{4(12)} \frac{1500}{4}x^{\frac{4}{3}}\,dx = \left(\frac{1500}{(3)(35)(10^6)(4)}\right)\left(\frac{3}{7}\right)(48)^{\frac{7}{3}}$$

$$\delta_A = 0.0128\text{ in.} \qquad \textbf{Ans.}$$

$$\delta = \int_0^L \frac{P(x)dx}{A(x)E}$$



$W = 500 \cdot x^{1/3}$ lb/in

$\frac{dP}{dx} = W$ (N/m ou lb/in)

$$P(x) = \int_0^x W\,dx = \int \underbrace{500 \cdot x^{1/3}}_{\text{lb/in}} \underbrace{dx}_{\text{in}} = 500 \frac{x^{4/3}}{\frac{4}{3}} = \underbrace{\frac{1500}{4} x^{4/3}}_{\text{lb}}$$

1 ft = 12 in

$$\delta = \int_0^{4\times 12} \underbrace{\frac{1500}{4} x^{4/3}}_{\text{lb}} \underbrace{dx}_{\text{in}} \cdot \frac{1}{\underbrace{3}_{\text{in}^2} \times \underbrace{35\ \text{Msi}}_{10^6 \text{ lb/in}^2}} = \frac{1500}{4 \times 3 \times 35 \cdot 10^6} \left[\frac{x^{7/3}}{7/3}\right]_0^{48}$$

$$= \frac{1500}{3 \times 35 \times 4 \cdot 10^6} \left(\frac{3}{7}\right) (48)^{7/3} = 0,0125\ \text{in}$$

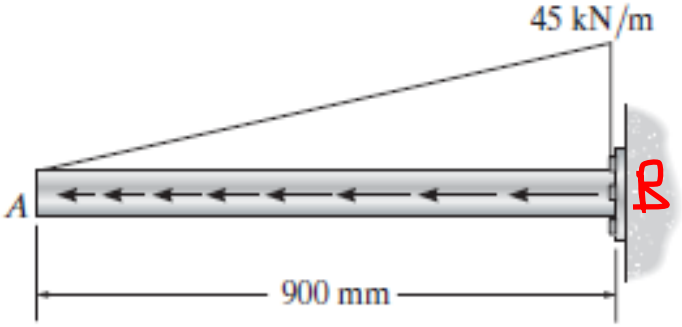
1 Msi = 10^6 lb/in²

1 ft = 12 in

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3. Barre avec déformation axiale non uniforme

F4–6. The 20-mm-diameter 2014-T6 aluminum rod is subjected to the triangular distributed axial load. Determine the displacement of end A.



F4–6

$y = ax + b$

$w(x)$

$\int w\,dx$

F4–6 Distributed load $P(x) = \dfrac{45(10^3)}{0.9}x = 50(10^3)x$ N/m

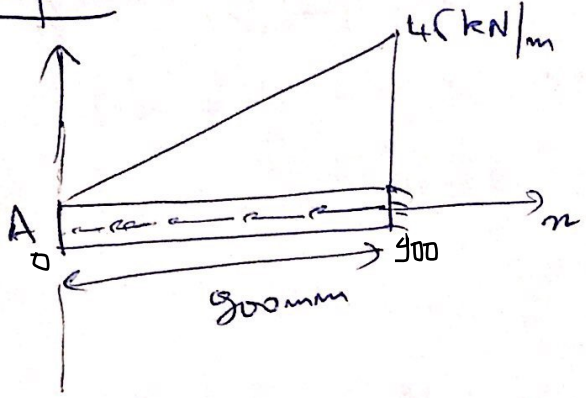
Internal load $P(x) = \frac{1}{2}(50(10^3))x(x) = 25(10^3)x^2$

$$\delta_A = \int_0^L \frac{P(x)dx}{AE}$$

$$= \frac{1}{[0.1(10^{-3})\pi][73.1(10^9)]}\int_0^{0.9\text{ m}} [25(10^3)x^2]dx$$

$$= 0.265\text{ mm}$$ *Ans.*

Exemple :



$W = ax + b$

$0 = a \cdot 0 + b \Rightarrow b = 0$

$45\,kN/m = a \times 900.10^{-3}\,m \Rightarrow a = \frac{45.10^3\,N/m}{900.10^{-3}\,m}$

$a = 50.10^3\ N/m^2$

$\boxed{W = 50.10^3 x \ (N/m)}$

$P(x) = \int_0^x W(u)\,du = \int_0^x 50.10^3 u\,du = 50.10^3 \left[\frac{u^2}{2}\right]_0^x = 50.10^3 \frac{x^2}{2}\ (N)$

$\delta = \int_0^{0,9} \frac{P(u)\,du}{A\,E} = \frac{1}{\left(\frac{20}{2}.10^{-3}\right)^2 \pi . 70.10^9} \int_0^{0,9} 50.10^3 \frac{u^2}{2}\,du$

$= \frac{25.10^3}{10^{-4} \pi . 70.10^9} \left[\frac{x^3}{3}\right]_0^{0,9} = 0,000\,278\ m = 0,278\ mm.$

ou après $E = 70.10^9$.

Si $E = 73,1.10^9 \Rightarrow \delta = 0,264\ mm.$

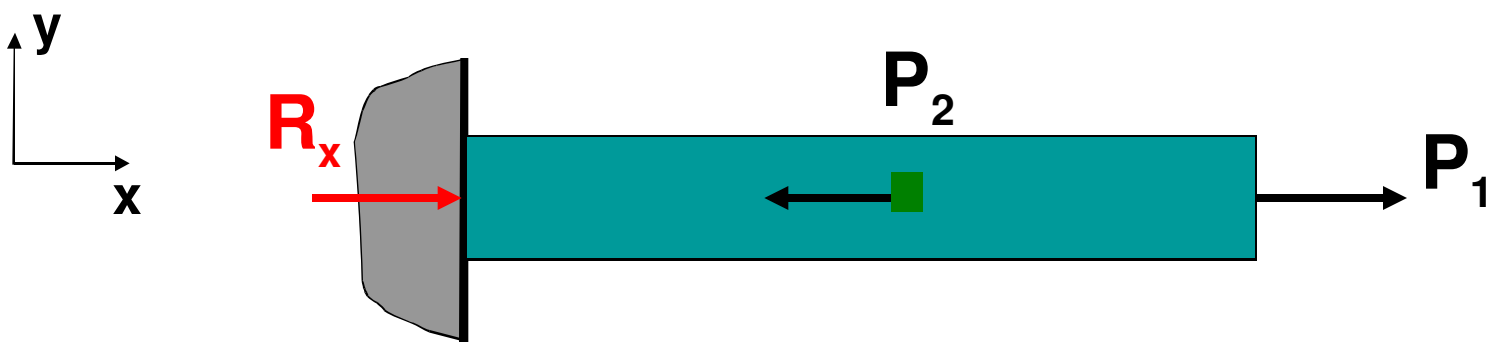
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4. Système hyperstatique

- **4.1 Qu'est-ce que c'est?**

Un système pour lequel les équations d'équilibre ne suffisent pas à calculer les réactions inconnues.

Exemple:

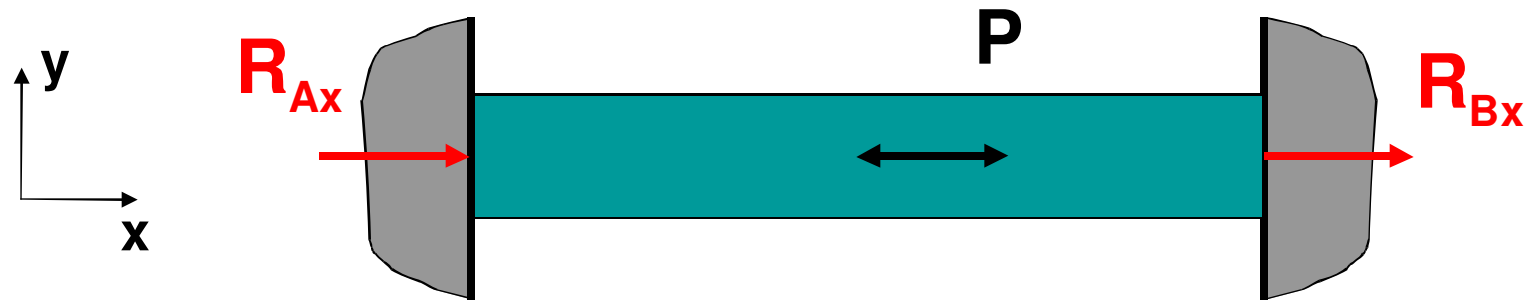


On a: $\sum F_x = 0 \Longrightarrow R_x = P_2 - P_1$

Ceci est un système isostatique

4. Système hyperstatique

Exemple:



Ceci est un système hyperstatique

Parce que R_{Ax} & R_{Bx} ne peuvent pas être calculées à partir des équations d'équilibre:

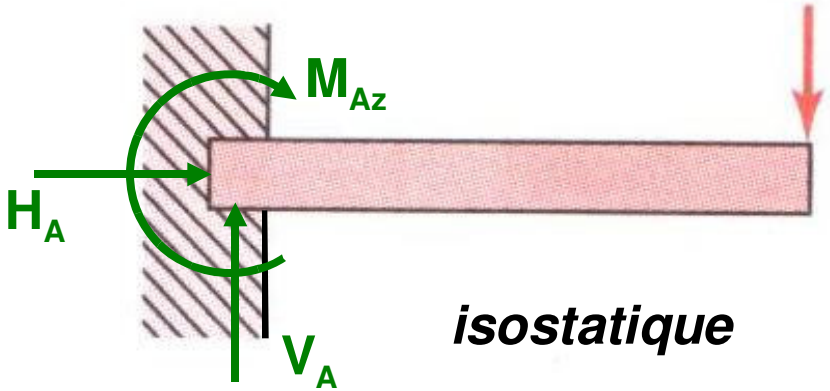
$$\sum F_x = 0 \quad \Rightarrow \quad R_{Ax} - P + R_{Bx} = 0$$

1 équation, 2 inconnues

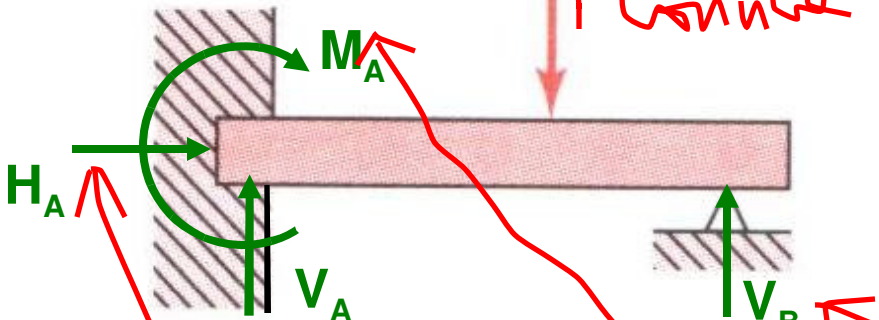
Par conséquent, il nous faut plus d'information!

4. Système hyperstatique

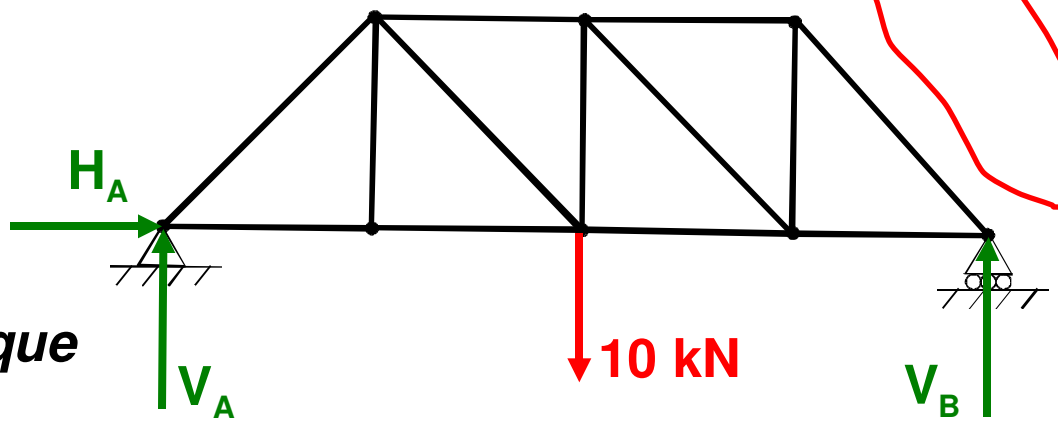
- Exemples additionnels:



isostatique

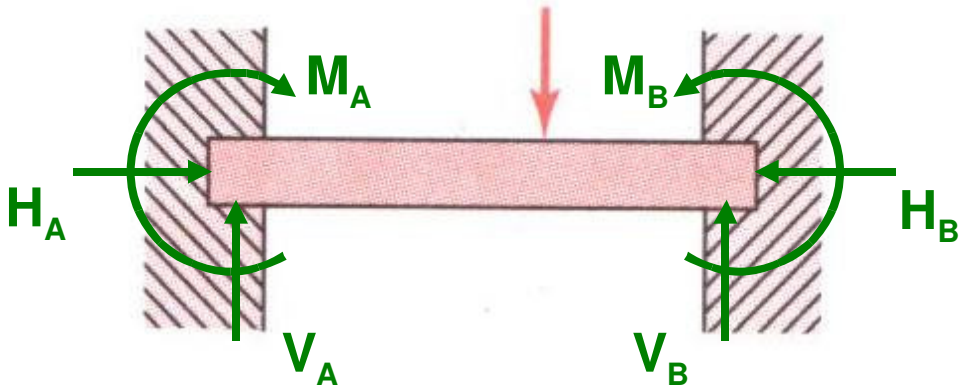


hyperstatique

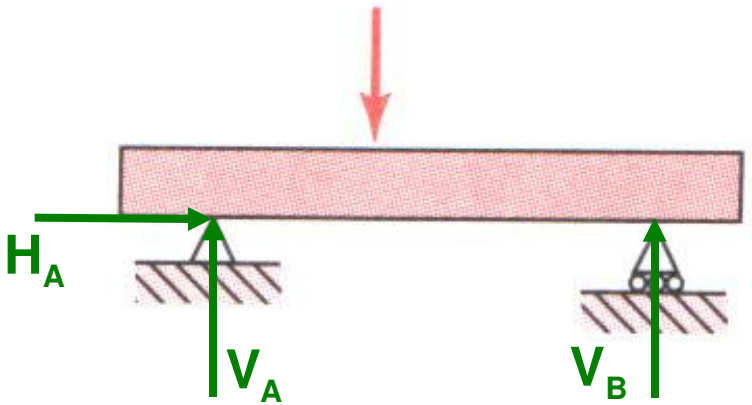


isostatique

4. Système hyperstatique



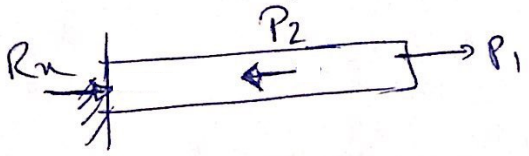
hyperstatique



isostatique

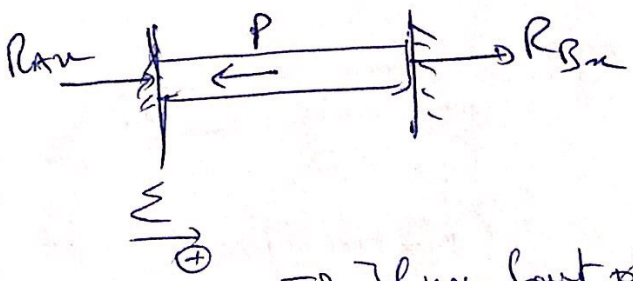
Système hyperstatique :

Nbre d'équations < Nbre d'inconnues.



$R_{Ax} + P_1 - P_2 = 0 \Rightarrow R_{Ax} = P_2 - P_1$

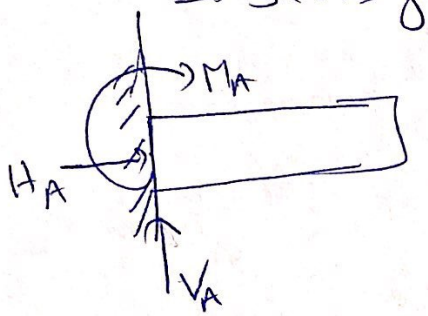
Système isostatique



$R_{Ax} + R_{Bx} - P = 0$

1 équation, 2 inconnues ! Syst. hyperstatique

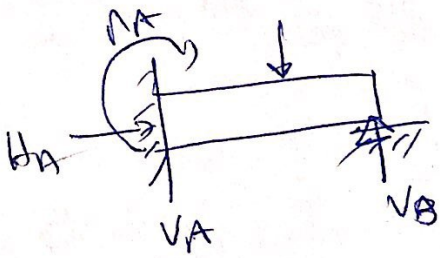
$\Rightarrow$ Il nous faut plus d'informations !



$\Sigma M = 0$
$\Sigma f_x = 0$
$\Sigma f_y = 0$

3 équations et 3 inconnues

Syst. isostatique



$\Sigma M = 0$
$\Sigma f_x = 0$
$\Sigma f_y = 0$

H_A, V_A, M_A, V_B

4 inconnues

et 3 équations

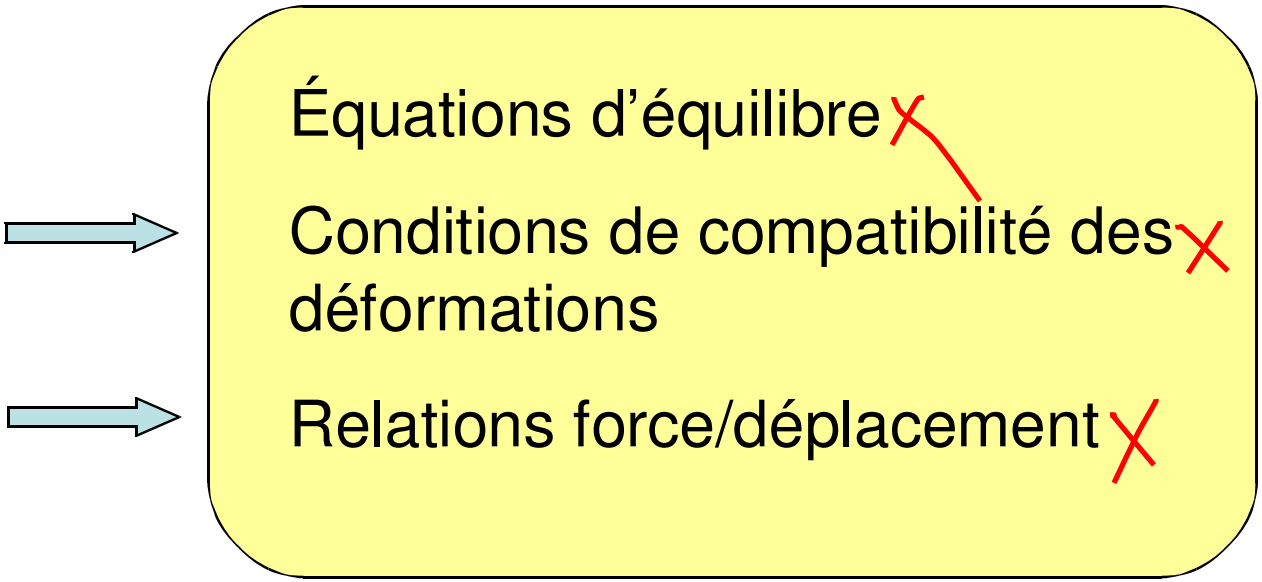
Système hyperstatique

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4. Système hyperstatique

4.2 Procédure d'analyse

Pour résoudre un système hyperstatique, il faut utiliser des équations supplémentaires:



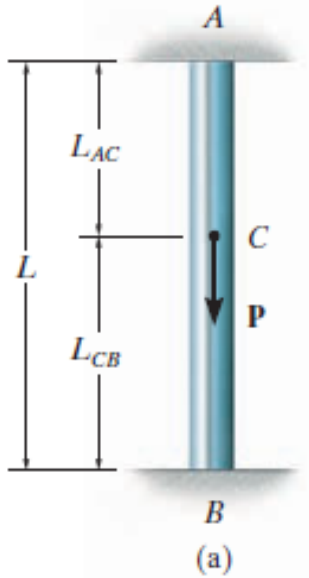
4. Système hyperstatique

$$+\uparrow \Sigma F = 0; \qquad F_B + F_A - P = 0$$

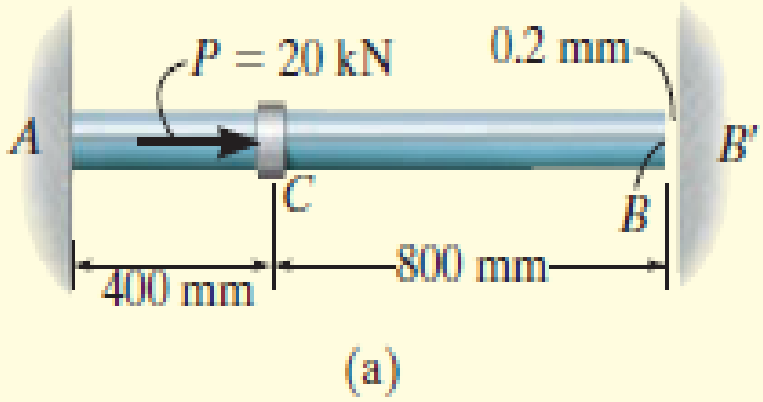
compatibility or *kinematic condition.* $\delta_{A/B} = 0$

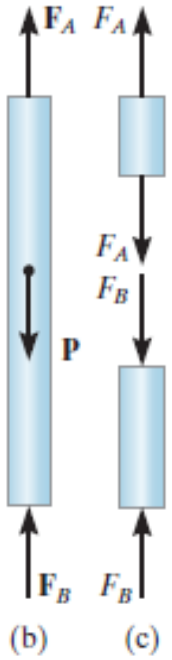
$$\frac{F_A L_{AC}}{AE} - \frac{F_B L_{CB}}{AE} = 0 \longrightarrow F_A = F_B(L_{CB}/L_{AC})$$

$$F_A = P\left(\frac{L_{CB}}{L}\right) \quad \text{and} \quad F_B = P\left(\frac{L_{AC}}{L}\right)$$



The steel rod shown in Fig. 4–12*a* has a diameter of 10 mm. It is fixed to the wall at *A*, and before it is loaded, there is a gap of 0.2 mm between the wall at *B*′ and the rod. Determine the reactions at *A* and *B*′ if the rod is subjected to an axial force of $P = 20$ kN as shown. Neglect the size of the collar at *C*. Take $E_{st} = 200$ GPa.





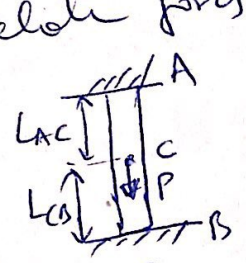
Résolution d'un système hyperstatique ?

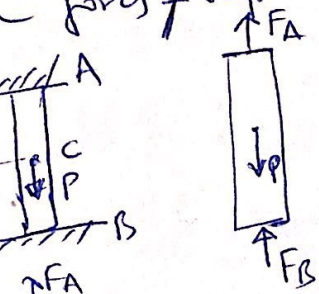
→ Equation d'équilibre $\sum \vec{F}_x = \vec{0}$ (ou $\sum \vec{F}_y = \vec{0}$)

→ Condition de compatibilité ($\delta = 0$ ou $\delta =$ valeur)

→ Relation force / déplacement ($\delta = \frac{PL}{AE}$)

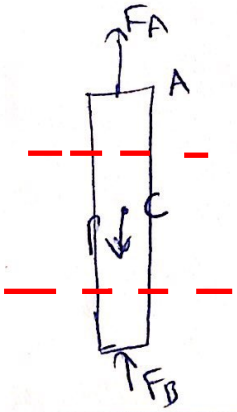
Exemple :

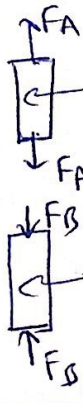




$F_A + F_B - P = 0$ (P donné)

F_A et F_B inconnues





$F_A \Rightarrow \delta_{AC} = \frac{F_A L_{AC}}{AE} > 0$ ($F_A > 0$)

$F_B \Rightarrow \delta_{CB} = \frac{-F_B}{AE} L_{BC} < 0$ ($F_B > 0$)

Cdt de compatibilité $\Rightarrow \delta_{A/B} = 0$

$\Rightarrow \delta_{A/B} = \delta_{A/C} + \delta_{C/B} = 0$

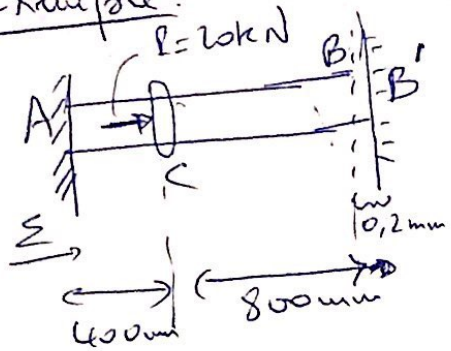
$\Rightarrow \delta_{AC} = -\delta_{CB} \Rightarrow \frac{F_A L_{AC}}{AE} = +\frac{F_B L_{BC}}{AE}$

$\boxed{F_A = \frac{F_B L_{BC}}{L_{AC}}}$ $\Rightarrow F_B \left[\frac{L_{BC}}{L_{AC}} + 1\right] = P$

$\Rightarrow F_B = \frac{P L_{AC}}{L}$, $F_A = \frac{P L_{BC}}{L}$

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Exemple :

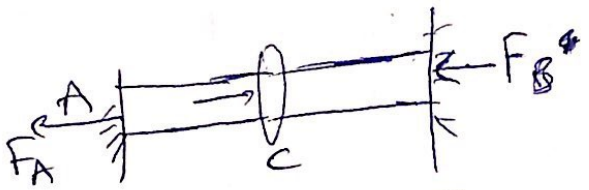


Acier : $E = 200\ GPa$

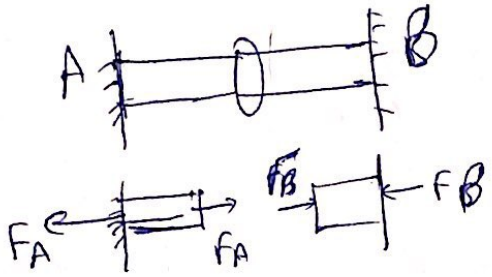
$d = 10\ mm$. A fixe.

Avant chargement on a un gap de 0,2 mm. entre B et B' (Mure)

$f_A = ?$ $f_B = ?$ (après chargement sur)



$\sum F_x = 0$ $\quad -F_A - F_B + P = 0 \Rightarrow \boxed{F_A + F_B = P}$



$\delta_{AC} = \dfrac{F_A L_{AC}}{AE}$, $\delta_{BC} = \dfrac{-F_B L_{BC}}{AE}$ si $(F_B \text{ si } > 0)$

Rque :

si B' est très loin $\Rightarrow F_B = 0$

$\boxed{f_A = 20\ kN = P}$

$F_B = 0$

$\hookrightarrow \delta_{AC} = \dfrac{F_A L_{AC}}{AE}$

$= 0,509\ mm$

$= \delta_{AB}$

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Condition de compatibilité :

$$\delta_{B/A} = 0,2\,mm$$

$$\Rightarrow \frac{F_A L_{AC} - F_B L_{BC}}{AE} = 0,2\,mm$$

$$F_A = \left(\left[(0,2.10^{-3}) (5.10^{-3})^2 \pi \times 200.10^9 \right] + F_B (800.10^{-3}) \right) \frac{1}{400.10^{-3}}$$

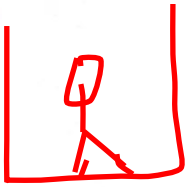
$F_A = 7853,98 + 2F_B$ et $F_A = P - F_B = 20.10^3 - F_B$

$\Rightarrow 7853,98 + 2F_B = 20000 - F_B$

$\Rightarrow F_B = 4,048\,kN$ et $F_A = 20 - 4,048 = 15,95\,kN$

$\Rightarrow \delta_{AC} = 0,406\,mm$; $\delta_{BC} = -0,206$

$\Rightarrow \delta_{AC} + \delta_{BC} = 0,2\,mm$



Note, Si on avait trouvé une valeur négative de F_B, c à d que le sens de F_B est vers la droite ($\rightarrow$), alors $\rightsquigarrow F_B = 0$ et $F_A = 20\,kN$, Cas isostatique !

Exemple, $BB' = 0,2\,cm \Rightarrow F_B < 0 \Rightarrow$ le sens supposé ($\leftarrow$) est faux ! $\Rightarrow$ pas de contact avec le mur !

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4. Système hyperstatique

Note:

Il existe d'autres méthodes pour résoudre les équations:

- La méthode de la force (Hibbeler, section 4.5)
- La méthode du déplacement

On ne verra pas ces méthodes dans le cadre du cours.

5. Contraintes thermiques

- Une variation de température entraine un allongement ou un raccourcissement du matériau.
- On obtient la déformation due à la dilatation thermique à l'aide de l'équation:

$$\varepsilon_T = \alpha \Delta T$$

où α est le coefficient de dilatation thermique ($^oC^{-1}$) et ΔT est le changement de température imposé (^oC)

- Pour une barre de longueur initiale L, l'allongement correspondant se calcule par:

$$\delta_T = \varepsilon_T L = \alpha L \Delta T$$

5. Contraintes thermiques

- Si la barre de longueur initiale L est aussi soumise à une force P, l'allongement causé par la force et le changement de température se calcule par:

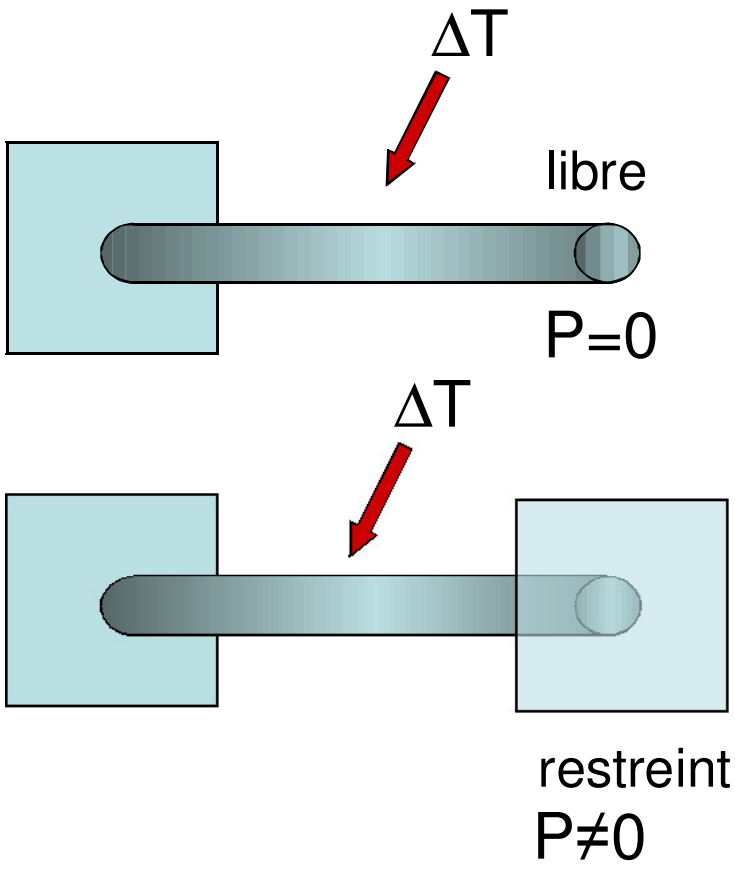
$$\delta = \frac{PL}{AE} + \alpha L \Delta T$$

- Note:
- Il n'y a pas de contrainte dans une barre qui peut se déformer librement sous l'effet d'un changement de température uniforme.
- Par contre si la barre est retenue, le changement de température induira une contrainte.

5. Contraintes thermiques

Exemples:

$$\delta = \frac{PL}{AE} + \alpha L \Delta T$$



$$\delta = \delta_T = \alpha L \Delta T$$

$$\delta = 0 = \frac{PL}{AE} + \alpha L \Delta T = 0$$

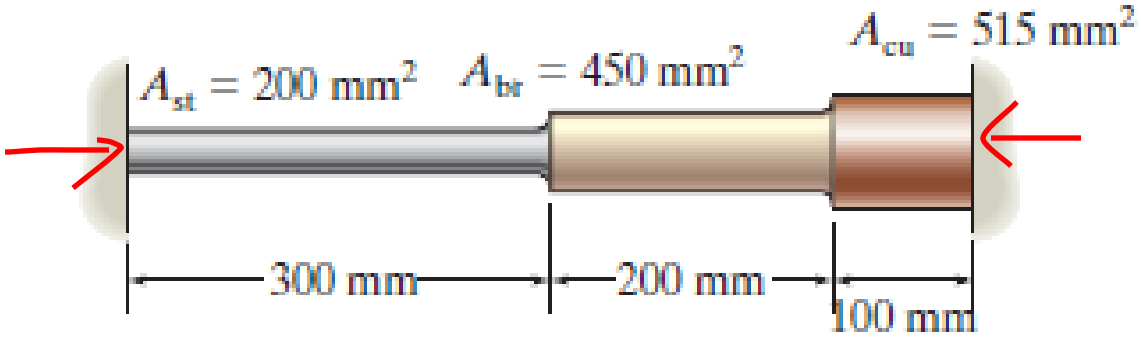
$$\frac{P}{A} = \sigma = -\alpha E \Delta T$$

5. Contraintes thermiques

•4–69. Three bars each made of different materials are connected together and placed between two walls when the temperature is $T_1 = 12°C$. Determine the force exerted on the (rigid) supports when the temperature becomes $T_2 = 18°C$. The material properties and cross-sectional area of each bar are given in the figure.

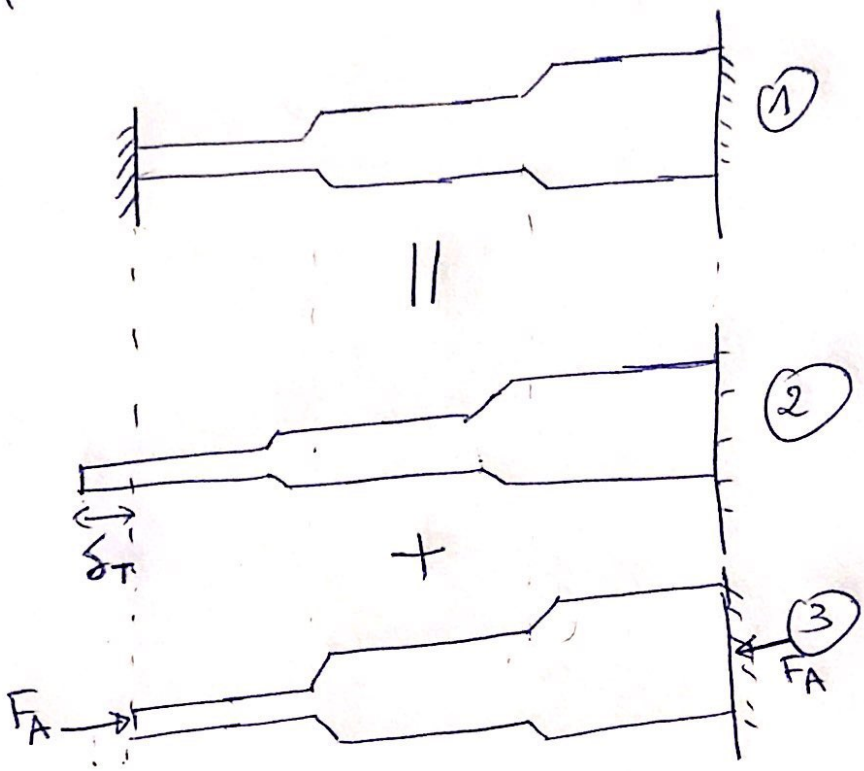
acier — Laiton — Cuivre

Steel	Brass	Copper
$E_{st} = 200$ GPa	$E_{br} = 100$ GPa	$E_{cu} = 120$ GPa
$\alpha_{st} = 12(10^{-6})/°C$	$\alpha_{br} = 21(10^{-6})/°C$	$\alpha_{cu} = 17(10^{-6})/°C$



Prob. 4–69

Exemple :



$F_A = ?$

(1) = (2) + (3)

$$\delta_{A/D} = \underbrace{\sum \delta_T}_{>0} + \underbrace{\sum \delta_F}_{<0}$$

$$= \sum \alpha_i L_i \Delta T_i - \sum \frac{F L_i}{A_i E_i} \quad ; \quad \Delta T_i = \Delta T \ \forall i \ (\text{fixé, Loi de variation, ...})$$

$\Delta T_i = \Delta T = 18 - 12$

$\Rightarrow \boxed{\Delta T = 6°C}$

or $\delta_{A/D} = 0 \Rightarrow \sum \delta_T = -\sum \delta_F$

$$\Rightarrow \sum \alpha_i L_i \Delta T = \sum \frac{F L_i}{A_i E_i}$$

$$\Rightarrow F = \frac{\sum \alpha_i L_i \Delta T}{\sum \frac{L_i}{A_i E_i}}$$

A.N :

$$F = \frac{12.10^{-6} \times 0,3 \times 6 + 21.10^{-6} \times 0,2 \times 6 + 17.10^{-6} \times 0,1 \times 6}{\frac{0,3}{200.10^{-6} \cdot 200.10^{3}} + \frac{0,2}{450.10^{-6} \cdot 100.10^{9}} + \frac{0,1}{515.10^{-6} \times 120.10^{9}}}$$

$= 4,2 kN$

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6. Conclusion

- On a étudié:

  - Comportement d'une barre soumise à des forces axiales
  - Les systèmes hyperstatiques
  - Comportement d'une barre soumise à une dilatation thermique
  - Des structures simples constitués de plusieurs barres